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FINITE PERTURBATION THEORY FOR NUCLEAR SPIN COUPLING CONSTANTS *

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Nuclear spin coupling constants are calculated using approximate SCF wavefunctions and a finite perturbation method. The results reproduce many of the experimental features and trends of coupling constants for carbon and hydrogen.

We are developing a simple theoretical method for studying those properties of molecules which involve polarization or distortion of the electron distribution (second order properties). Basically, the technique is to calculate wave functions for a molecule in the presence of a finite perturbation and then compare details of the electron density with the corresponding density for the unperturbed system. A closely related method has already been used by Cohen and Roothaan [1-3] in calculations of the electrical polarizabilities of atoms and by Schweig [4,5] for polarizabilities of certain π electron systems. We are using the method for calculations of electrical polarizabilities, magnetic susceptibilities, NMR chemical shifts and nuclear spin coupling constants. In this communication, we present some encouraging preliminary results for spin coupling constants.

The theory of nuclear spin-spin coupling through the Fermi contact mechanism is easily developed along the following lines. If two nuclei A, B are treated as fixed magnetic dipoles μ_A, μ_B oriented in the z-direction, then the complete Hamiltonian can be written as

$$\mathcal{H}(\mu_A, \mu_B) = \mathcal{H}^{(0)} + \mu_A \mathcal{H}^{(1)}(A) + \mu_B \mathcal{H}^{(1)}(B) . \quad (1)$$

Here $\mathcal{H}^{(1)}(A)$ is the Hamiltonian representing the contact interaction between the electron moments and unit nuclear moment in the z-direction,

$$\mathcal{H}^{(1)}(A) = \tfrac{8}{3}\,\beta \sum_k \delta(r_{kA}) S_{kz} , \quad (2)$$

where β is the Bohr magneton and S_{kz} the z-component of the spin of electron k. Using the Hellman-Feynmann theorem [6,7], it can be shown that the reduced coupling constant K_{AB} (that is, the coupling constant per unit magnetic moments) [8] is

$$K_{AB} = \frac{\partial^2 E(\mu_A, \mu_B)}{\partial\mu_A \partial\mu_B} = \frac{\partial}{\partial\mu_B} \times$$
$$\times \langle \psi(0, \mu_B) | \mathcal{H}^{(1)}(A) | \psi(0, \mu_B) \rangle , \quad (3)$$

the derivatives to be evaluated at $\mu_A = \mu_B = 0$. Although the Hellman-Feynmann theorem is normally valid only for exact or fully optimized wavefunctions, it can be demonstrated to also hold true for approximate SCF wavefunctions if the basis is independent of the perturbation.

The quantity $\langle \psi(0, \mu_B | \mathcal{H}^{(1)}(A) | \psi(0, \mu_B) \rangle$ to be differentiated in (3) is proportional to the electron spin density at nucleus A when a perturbing magnetic dipole μ_B is placed at nucleus B. In the finite perturbation approach, we evaluate this quantity for finite μ_B.

In the calculations reported here, the wavefunction $\psi(0, \mu_B)$ was an unrestricted single determinant molecular orbital function (different orbitals for different spins) of the INDO (intermediate neglect of differential overlap) type recently developed here [9]. (This method uses only valence atomic orbitals and neglects diatomic differential overlap in all two-electron integrals except for the one-center exchange type). Within the framework of the approximations of the INDO method, the perturbation $\mu_B \mathcal{H}^{(1)}(B)$ may be introduced to the one-electron (core) Hamiltonian for α orbitals by adding a small quantity h to the diagonal matrix element representing the valence s-orbital of atom B. At the same time, $-h$ is

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Table 1
Calculated and experimental values of coupling constants J (in c/s).

	Calc.	Exp.	Ref.
Hydrogen H-H	408.65	280	[11]
Methane H-H	-6.13	-12.4	[12]
Ethane Gemical H-H	-5.22		
Methyl Fluoride H-H	-1.86	-9.6	[13]
Ethylene Gemical H-H	3.24	2.5	[14]
Formaldehyde H-H	31.86	40.2	[15]
Water H-H	-8.07	-7.2	[16]
Ethane Gauche Vicinal H-H	3.25		
Ethane Trans Vicinal H-H	18.63		
Ethane Vicinal H-H (Average)	8.37	8	[14]
Ethylene Cis Vicinal H-H	9.31	11.7	[14]
Ethylene Trans Vicinal H-H	25.15	19.1	[14]
Acetylene H-H	10.99	9.5	[14]
Benzene Ortho H-H	8.15	7.54	[17]
Benzene Meta H-H	2.13	1.37	[17]
Benzene Para H-H	1.15	0.69	[17]
Methane C-H	122.92	125	[18]
Ethane Bonded C-H	122.12	124.9	[14]
Ethylene Bonded C-H	156.71	156.4	[14]
Acetylene Bonded C-H	232.66	248.7	[14]
Benzene Bonded C-H	140.29	157.5	[19]
Ethane Long Range C-H	-7.20	-4.5	[14]
Ethylene Long Range C-H	-11.57	-2.4	[14]
Acetylene Long Range C-H	2.52	49.3	[14]
Benzene Ortho C-H	-4.94	1.0	[19]
Benzene Meta C-H	9.40	7.4	[19]
Benzene Para C-H	-2.27	-1.1	[19]
Ethane C-C	41.45	34.6	[14]
Ethylene C-C	82.14	67.6	[14]
Acetylene C-C	163.74	171.5	[14]

added to the corresponding β element. The spin density at nucleus A is then taken to be proportional to the difference between the α and β populations of the valence s orbital of A($\rho_{s_A s_A}$). The required derivative with respect to h was then evaluated as $\rho_{s_A s_A}(h)/h$. For sufficiently small h this becomes equal to the derivative at $h = 0$. The only remaining quantities required to relate this derivative to the coupling constants are then the densities of the atomic s-orbitals at the nuclei.

The results for H-H, C-H and C-C coupling constants (in c/s) for a few simple molecules (using standard geometries) [10] are given in table 1. The atomic s-orbital densities at the nuclei for C and H were chosen to obtain optimum overall fit with the data (by minimizing the sum of the squares of the differences between calculated and experimental K-values).

As can be seen from the table, many well-established experimental trends and features are clearly reproduced by the theory. Thus gemical H-H constants are negative for sp^3 carbon, but positive for sp^2 in ethylene and large and positive for formaldehyde. Vicinal H-H constants are always positive and greatest in the trans configuration. The longer range meta and para H-H constants in benzene are positive. Directly bonded C-H constants increase along the series *ethane, ethylene, acetylene*, as do the C-C *constants*. The long range C-C-H constants are less well reproduced, but the small benzene value is the only one for which the calculated sign is apparently incorrect.

Fuller details of the theory and more extensive results will be given in a forthcoming publication.

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